



# BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - I - Apr - 2026

II Semester

## 22CH203 & ENGINEERING CHEMISTRY II

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| 1 | (i)  | <p>A student assembles a half-cell using a metal electrode dipped in an electrolyte but is unable to measure its absolute potential directly. The student then connects it to a standard hydrogen electrode to obtain a measurable value. The setup produces a stable voltage reading.</p> <p>Analyze why a reference electrode is necessary and determine how it enables measurement of electrode potential.</p> <p>(3 Marks - [An/C, 2])</p> |
| 2 | (ii) | <p>A closed electrochemical circuit shows continuous electron flow through a wire while ions move through the electrolyte. The system maintains neutrality despite ongoing reactions. Determine how ionic movement supports sustained electron flow in the circuit.</p> <p>(2 Marks - [U/C, 2])</p>                                                                                                                                            |
| 3 | (i)  | <p>An electrolyte solution under an applied electric field shows that cations move faster than anions due to smaller size and higher mobility, contributing unevenly to current flow.</p> <p>Analyze how transport numbers arise in this system and determine their significance in current distribution.</p> <p>(3 Marks - [An/C, 2])</p>                                                                                                     |
| 4 | (ii) | <p>In an electrolyte under steady conditions, both electric field and concentration gradient are present, influencing ion flux simultaneously.</p> <p>Interpret how the Nernst-Planck framework integrates these effects to govern ion movement.</p> <p>(2 Marks - [Ap/C, 2])</p>                                                                                                                                                              |
| 5 | (i)  | <p>In an experiment, a metal electrode is immersed in its ionic solution, and a potential develops at the interface. The system shows no external current flow but still maintains a measurable potential difference across the interface.</p> <p>Interpret the origin of this interfacial potential and infer the role of equilibrium in its formation.</p> <p>(3 Marks - [Ap/C, 2])</p>                                                      |
| 6 | (ii) | <p>Two different metals are connected without an electrolyte, and no current is observed despite a difference in their properties.</p> <p>Infer why an electrolyte is essential for current generation in electrochemical systems.</p> <p>(2 Marks - [U/C, 2])</p>                                                                                                                                                                             |

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| 7  | (i)  | <p>A battery exhibits increased internal resistance as it ages, even though electrode materials remain unchanged, and electrolyte conductivity slightly decreases over time.</p> <p>Evaluate the factors contributing to increased internal resistance and infer its effect on battery performance.</p> <p>(3 Marks - [Ap/C, 2])</p>                                                                                                                                                                                                                                                                                                                                                                                         |
| 8  | (ii) | <p>An electrochemical system experiences enhanced ion transport when stirred externally, even though no changes are made to electric field or concentration.</p> <p>Determine the role of convection in this system and justify its effect on transport enhancement.</p> <p>(2 Marks - [U/C, 2])</p>                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 9  | (i)  | <p>An industrial electrochemical setup operates under varying environmental conditions where temperature increases gradually and reactant concentration changes due to continuous consumption, while pressure variations also occur in a gas electrode system, leading to fluctuations in measured cell potential over time.</p> <p>Determine how variations in concentration, temperature, and pressure influence the observed EMF and justify the combined effect of these parameters using electrochemical reasoning.</p> <p>(5 Marks - [U/C, 2])</p>                                                                                                                                                                     |
| 10 | (i)  | <p>An industrial system uses an electrolytic cell to deposit a metal coating onto a conductive surface. The setup includes an inert platinum electrode as the anode and a metal ion solution. Operators notice uneven deposition and increased energy consumption during prolonged operation.</p> <p>Evaluate the operational inefficiencies in this electrolytic cell by correlating electrode type, reaction pathways, and energy requirements, and recommend how system design influences performance outcomes.</p> <p>(5 Marks - [Ap/C, 2])</p>                                                                                                                                                                          |
| 11 | (i)  | <p>A researcher evaluates two electrochemical cells, one designed to operate reversibly under controlled slow reactions and another subjected to rapid discharge with significant energy losses, and compares their efficiencies while attempting to relate electrical output to thermodynamic feasibility.</p> <p>Analyze the differences between the two cells in terms of reversibility and infer how these differences affect the relationship between EMF and Gibbs free energy.</p> <p>(5 Marks - [An/C, 2])</p>                                                                                                                                                                                                       |
| 12 | (i)  | <p>A laboratory team constructs an electrochemical setup using a zinc rod immersed in zinc sulfate solution and a copper rod immersed in copper sulfate solution, connected via a salt bridge containing potassium nitrate. During operation, they observe electron flow through the external circuit and ion migration through the salt bridge, but also note a gradual decrease in voltage over time due to changes in ion concentration near the junction.</p> <p>Analyze the working mechanism of this electrochemical cell by determining the role of each component and infer the reasons for voltage variation with respect to junction behavior and ion movement within the system.</p> <p>(5 Marks - [Ap/C, 2])</p> |

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| 13 | (i)  | <p>A metal electrode is immersed in an aqueous electrolyte containing both cations and anions, where the metal surface acquires a net charge due to electron transfer. The nearby ions rearrange themselves forming a structured region near the interface. The compactness and distribution of ions vary depending on thermal motion and electrostatic attraction. Experimental observations indicate that the simple rigid-layer assumption fails to explain the diffuse nature of ion distribution at higher temperatures.</p> <p>Analyze how different theoretical models interpret the structure of the electrical double layer at the electrode–electrolyte interface and justify which model best reconciles both compact and diffuse charge distributions under varying conditions.</p> <p>(5 Marks - [An/C, 2])</p> |
| 14 | (i)  | <p>An electrochemical cell operates at 298 K where the EMF is measured as 0.59 V for a reaction involving transfer of 2 electrons. The student is required to determine the equilibrium constant for the reaction using thermodynamic relationships linking EMF and equilibrium.</p> <p>Formulate the relationship between EMF and equilibrium constant and determine the magnitude of the equilibrium constant using the provided data.</p> <p>(5 Marks - [Ap/C, 3])</p>                                                                                                                                                                                                                                                                                                                                                    |
| 15 | (i)  | <p>An electrochemical cell shows variation in interfacial potential when the concentration of electrolyte changes. Measurements reveal that the potential drop is not linear but varies significantly close to the electrode surface. The electric field strength is highest near the electrode and gradually decreases into the bulk solution.</p> <p>Determine how the potential gradient develops across the electrical double layer and evaluate its relationship with charge distribution and capacitance at the interface.</p> <p>(5 Marks - [U/C, 2])</p>                                                                                                                                                                                                                                                             |
| 16 | (i)  | <p>A concentration cell is built using the same metal electrode but with ion concentrations of 0.01 M and 1 M respectively. The system operates at 298 K, and the student uses the Nernst equation to estimate the cell potential, considering logarithmic dependence on ion concentration.</p> <p>Determine how the difference in concentration influences the EMF and evaluate the role of the reaction quotient in predicting the direction of the cell operation.</p> <p>(5 Marks - [U/C, 3])</p>                                                                                                                                                                                                                                                                                                                        |
| 17 | (i)  | <p>Different electrode materials are tested under standard conditions. Each combination produces a different EMF. The goal is to identify the most efficient pair.</p> <p>Determine how standard electrode potentials influence EMF and energy efficiency.</p> <p>(3 Marks - [Ap/C, 2])</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 18 | (ii) | <p>A system uses measured EMF to estimate electrical energy output. The cell operates under known conditions. The output depends on voltage.</p> <p>Determine how EMF relates to electrical energy output of the cell.</p> <p>(2 Marks - [U/C, 1])</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

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| 19 | (i)  | <p>An industrial battery system operates continuously and shows a sequence of losses starting from activation barriers to concentration imbalances and finally resistive heating under load conditions.</p> <p>Formulate the logical sequence of processes that convert an ideal electrochemical cell into an irreversible one during operation.</p> <p>(3 Marks - [U/C, 1])</p>                             |
| 20 | (ii) | <p>A comparison between two cells shows one maintaining constant voltage under ideal conditions while another shows fluctuating voltage and energy loss during operation under real-world conditions.</p> <p>Analyze how operational behavior differentiates reversible and irreversible cells in terms of energy efficiency.</p> <p>(2 Marks - [Ap/C, 1])</p>                                               |
| 21 | (i)  | <p>An electrochemical system operates at constant temperature but varying electrolyte concentrations. The EMF changes with concentration differences. The energy output becomes variable.</p> <p>Analyze how concentration changes affect EMF and infer the impact on energy output.</p> <p>(3 Marks - [An/C, 2])</p>                                                                                        |
| 22 | (ii) | <p>A cell shows reduced output at higher temperature compared to optimal conditions. The system deviates slightly from equilibrium. The energy produced decreases.</p> <p>Infer how temperature variation affects Gibbs free energy and EMF.</p> <p>(2 Marks - [Ap/P, 2])</p>                                                                                                                                |
| 23 | (i)  | <p>A laboratory cell operating under ideal conditions is compared with a real system where entropy production is observed due to energy dissipation and heat generation during operation, especially under sustained current flow.</p> <p>Integrate entropy production with energy loss mechanisms to infer why practical efficiency deviates from theoretical predictions.</p> <p>(3 Marks - [An/P, 2])</p> |
| 24 | (ii) | <p>A small electronic device battery shows slight heating and reduced output voltage when used for extended durations, indicating minor but noticeable inefficiencies in energy delivery.</p> <p>Determine how such small inefficiencies indicate irreversible behavior and infer their effect on device performance.</p> <p>(2 Marks - [An/C, 2])</p>                                                       |

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| 25 | (i) | <p>A remote healthcare unit in a mountainous region must select an electrochemical energy source to power diagnostic equipment continuously without frequent maintenance access. The team evaluates primary and secondary cells considering rechargeability, reaction reversibility, and environmental stability. Constraints include limited transportation, fluctuating temperatures, and the need for consistent energy output over long durations. Engineers analyze energy density, efficiency, and electrode durability to ensure reliable and sustainable operation.</p> <p>Determine the most suitable electrochemical cell classification for the healthcare unit by integrating energy storage requirements, reaction reversibility, and operational constraints, and justify the decision through systematic reasoning.</p> <p>(5 Marks - [Ap/C, 3])</p>                                                                                                                                                                              |
| 26 | (i) | <p>A renewable energy storage system in a remote microgrid relies on a bank of lithium-ion batteries that undergo frequent charge and discharge cycles due to intermittent solar input and fluctuating load demand across the day. The system operator observes that although the batteries initially delivered expected energy output, over time the usable capacity has declined and charging requires longer durations under similar conditions. Temperature variations between day and night are significant, and the depth of discharge varies depending on energy demand peaks. The operator must redesign the charging strategy to maintain reversibility and prolong system efficiency under these fluctuating environmental and operational conditions.</p> <p>Analyze the given system behavior and determine an optimized charging and discharging strategy that preserves electrode reversibility while minimizing efficiency losses, integrating relevant electrochemical and operational factors.</p> <p>(5 Marks - [An/C, 3])</p> |
| 27 | (i) | <p>A manufacturing plant observes inconsistent voltage output and rapid efficiency loss in electrochemical cells used for industrial sensors. The cells contain metal electrodes, liquid electrolytes, and separators, but performance varies across batches. Engineers suspect issues such as electrode degradation, electrolyte contamination, or separator malfunction affecting ion flow. Environmental exposure and assembly inconsistencies may also influence redox reactions and energy output.</p> <p>Analyze the possible causes of performance degradation by correlating component-level failures with electrochemical principles and infer the most critical fault affecting energy output.</p> <p>(5 Marks - [U/C, 3])</p>                                                                                                                                                                                                                                                                                                         |
| 28 | (i) | <p>An electric vehicle manufacturer compares two battery management strategies: one prioritizes full discharge to maximize range, while the other restricts discharge to 60% capacity to enhance battery lifespan. Field data shows that vehicles using full discharge deliver higher immediate performance but require battery replacement sooner, whereas restricted discharge vehicles show longer cycle life but reduced usable energy per cycle. The company must determine the optimal trade-off between performance and longevity.</p> <p>Analyze the trade-offs involved in both strategies and recommend an optimal approach by evaluating efficiency, cycle life, and electrochemical reversibility considerations.</p> <p>(5 Marks - [An/C, 3])</p>                                                                                                                                                                                                                                                                                   |

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| 29 | (i) | <p>A research laboratory is testing an electrochemical cell where both electrodes are maintained under identical temperature and pressure while allowing ion exchange through a salt bridge. The system is allowed to operate very slowly so that equilibrium is nearly maintained at all times. The engineers observe no heat loss and maximum electrical work output during operation. However, they are unsure whether the cell fully satisfies thermodynamic reversibility conditions.</p> <p>Analyze the given system and determine whether the electrochemical cell qualifies as reversible by evaluating the thermodynamic conditions and operational characteristics. Justify your reasoning by integrating equilibrium behavior and energy considerations.</p> <p>(5 Marks - [An/C, 3])</p> |
| 30 | (i) | <p>A national electric grid operator is planning large-scale integration of renewable energy sources such as solar and wind, but faces inconsistencies in supply due to intermittency and storage inefficiencies. Existing lithium-ion battery systems are deployed but show limitations in energy density and cost, leading to high infrastructure expenses and reduced feasibility for long-term storage. Engineers are evaluating alternative storage solutions while considering scalability and lifecycle performance under fluctuating load demands.</p> <p>Analyze the constraints imposed by current electrochemical storage systems in the given scenario and determine how these limitations influence large-scale renewable integration decisions.</p> <p>(5 Marks - [Ap/C, 3])</p>       |
| 31 | (i) | <p>A Daniell cell is constructed using Zn and Cu electrodes with their respective sulfate solutions. The concentration of <math>\text{Zn}^{2+}</math> ions is gradually increased while <math>\text{Cu}^{2+}</math> concentration remains constant. The experiment is conducted under controlled conditions, and the potential difference is continuously monitored.</p> <p>Determine how the change in <math>\text{Zn}^{2+}</math> concentration influences the cell potential using appropriate electrochemical principles, and justify the direction of variation by integrating equilibrium concepts.</p> <p>(5 Marks - [U/P, 3])</p>                                                                                                                                                            |
| 32 | (i) | <p>A research lab is comparing solid-state batteries with conventional lithium-ion batteries for use in aerospace applications. The focus is on safety, energy density, and operational reliability under extreme temperature variations and mechanical stress conditions.</p> <p>Formulate a comparative assessment and infer how solid-state technology addresses the limitations of traditional systems in this context.</p> <p>(5 Marks - [An/C, 3])</p>                                                                                                                                                                                                                                                                                                                                         |
| 33 | (i) | <p>A buried pipeline experiences corrosion at certain sections where soil moisture and composition vary. Investigations reveal that some regions act differently despite being part of the same metal structure.</p> <p>Determine how variations in environmental conditions lead to formation of corrosion cells within the same metal and infer the roles of different regions in the process.</p> <p>(3 Marks - [E/C, 2])</p>                                                                                                                                                                                                                                                                                                                                                                     |

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| 34 | (ii) | <p>During a laboratory experiment, a metal strip is partially immersed in an electrolyte solution, and corrosion is observed only in the immersed region.</p> <p>Infer the essential components required for corrosion to occur in this setup and justify their interaction.</p> <p>(2 Marks - [An/C, 2])</p>                                                                                                                                                                                        |
| 35 | (i)  | <p>A corrosion engineer studies two metals exposed separately to humid air and dry air at room temperature. The metal in humid air corrodes rapidly, while the one in dry air shows negligible corrosion. The engineer suspects the presence of an electrolyte film is critical for electron transfer continuity.</p> <p>Interpret how the presence of moisture alters electron transfer pathways and justify its impact on corrosion rate differences.</p> <p>(3 Marks - [An/C, 2])</p>             |
| 36 | (ii) | <p>Two identical metal rods are placed in different solutions: one neutral and one strongly acidic. The acidic solution shows rapid corrosion with gas evolution.</p> <p>Infer the role of electrolyte composition in modifying electron transfer reactions and corrosion intensity.</p> <p>(2 Marks - [An/C, 1])</p>                                                                                                                                                                                |
| 37 | (i)  | <p>A metallic bridge exposed to coastal conditions shows uniform rusting across large areas. Engineers classify this as a common type of corrosion influenced by environmental exposure.</p> <p>Analyze the type of corrosion occurring and determine the contributing factors that lead to its uniform nature.</p> <p>(3 Marks - [An/C, 2])</p>                                                                                                                                                     |
| 38 | (ii) | <p>A student observes that when zinc reacts in a solution, it loses electrons while another species gains electrons.</p> <p>Determine the distinction between oxidation and reduction processes in this context and relate it to corrosion.</p> <p>(2 Marks - [An/C, 2])</p>                                                                                                                                                                                                                         |
| 39 | (i)  | <p>A protective coating on a metal surface develops microscopic defects, allowing small regions to be exposed to a corrosive electrolyte. These exposed areas begin to corrode faster than the surrounding protected surface. Engineers analyze the system to understand how electron flow is concentrated in these regions.</p> <p>Analyze how localized defects influence electron transfer distribution and infer their role in accelerating corrosion kinetics.</p> <p>(3 Marks - [An/C, 2])</p> |
| 40 | (ii) | <p>A metal sample is heated in an oxygen-rich furnace, and a uniform oxide layer forms across its surface. The growth rate slows down over time. Engineers consider diffusion and electron transfer limitations.</p> <p>Determine how electron transfer contributes to the decreasing rate of oxidation over time.</p> <p>(2 Marks - [E/C, 1])</p>                                                                                                                                                   |

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| 41 | (i) | <p>A steel pipeline buried underground shows uneven corrosion patterns, with certain regions dissolving faster than others despite being made of the same material. Soil moisture, oxygen availability, and ionic content vary along the pipeline, leading to localized electrochemical activity. Investigations reveal formation of anodic and cathodic regions due to environmental heterogeneity, and measurable potential differences exist across segments of the metal surface.</p> <p>Analyze the electrochemical basis for the formation of anodic and cathodic regions in the given system and determine how these localized regions drive corrosion progression across the pipeline.</p> <p>(5 Marks - [An/C, 3])</p>                                                                                                                  |
| 42 | (i) | <p>A pipeline system transporting saline water incorporates stainless steel valves connected to carbon steel pipes, where protective coatings are applied but periodically damaged due to mechanical wear. Field data indicates accelerated corrosion near the connection points, especially when flow stagnation occurs. Engineers suspect that both material differences and environmental factors are contributing to failure.</p> <p>Determine the governing factors leading to corrosion in this system and recommend how the galvanic series and environmental control can be used to mitigate the issue.</p> <p>(5 Marks - [E/C, 3])</p>                                                                                                                                                                                                  |
| 43 | (i) | <p>Two metals, zinc and copper, are placed in an electrolyte solution and electrically connected. Standard electrode potentials indicate zinc has a more negative potential compared to copper. After some time, zinc shows significant material loss while copper remains largely unaffected.</p> <p>Evaluate the role of standard electrode potentials in predicting corrosion tendency and justify the observed behavior of the metals in this system.</p> <p>(5 Marks - [E/C, 3])</p>                                                                                                                                                                                                                                                                                                                                                        |
| 44 | (i) | <p>A coastal engineering firm is selecting materials for a pier structure exposed to seawater, where steel piles are connected to bronze fittings and partially coated with protective paint that has degraded unevenly over time. The seawater acts as a conductive electrolyte, and inspection reports indicate localized metal loss near junction regions where two metals meet. The engineers also observe that larger steel areas are connected to smaller bronze components, and the dissolved oxygen concentration varies across the structure due to tidal movement.</p> <p>Analyze the electrochemical interactions within the given system and determine which metal will preferentially corrode, while justifying how the galvanic series and area ratio influence the observed degradation pattern.</p> <p>(5 Marks - [An/C, 3])</p> |
| 45 | (i) | <p>In a semiconductor device, copper interconnects are subjected to alternating current conditions. Although current direction changes, gradual degradation is still observed. Material transport is influenced by both concentration gradients and transient electric fields. Device performance declines due to structural instability at microscopic scales.</p> <p>Formulate how multiple mass transport mechanisms collectively contribute to degradation under alternating current conditions and justify their relative significance.</p> <p>(5 Marks - [An/C, 3])</p>                                                                                                                                                                                                                                                                    |
| 46 | (i) | <p>A ship hull operating continuously in seawater is protected using zinc blocks attached to its surface. Over time, the zinc blocks show significant material loss while the steel structure remains intact.</p> <p>Interpret the electrochemical mechanism that enables zinc to protect the steel hull and determine how material properties influence the effectiveness of this corrosion prevention strategy.</p> <p>(5 Marks - [An/C, 3])</p>                                                                                                                                                                                                                                                                                                                                                                                               |



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| 47 | (i) | <p>A buried pipeline carrying conductive fluid is exposed to stray electrical currents from nearby power systems. Over time, localized corrosion spots appear where current enters and exits the metal surface. Ionic species in the surrounding soil show directional movement aligned with potential gradients. The corrosion rate varies across different regions of the pipeline.</p> <p>Determine how ionic transport mechanisms interact with electric potential gradients to influence spatial variation in corrosion intensity along the pipeline.</p> <p>(5 Marks - [Ap/C, 3])</p>                                                                                                      |
| 48 | (i) | <p>An underground steel pipeline transporting crude oil across a coastal region is experiencing accelerated corrosion due to saline soil conditions. Engineers propose using a cathodic protection system but must decide between sacrificial anode and impressed current methods while considering installation cost, maintenance frequency, and effectiveness over long distances.</p> <p>Analyze the suitability of selecting either sacrificial anode or impressed current cathodic protection for this pipeline by evaluating operational constraints, and justify the most effective method based on corrosion control efficiency and system scalability.</p> <p>(5 Marks - [An/C, 3])</p> |